Pronounced deviation from healthy gait ankle kinematics leads to an increase in knee joint loads

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Introduction

When walking, the knee joint itself endures forces that account between 2 and 5 times the body-weight [1]. A relation between the knee and ankle kinematics can be inferred by changing the Achilles tendon stiffness [2] and gait adaptations can be produced due to plantarflexor weakness and contraction [3]. In general, studies show how changes in specific ankle kinematic scenarios affect those of the knee, but not their direct consequences on the load. We look to find a correlation between how differences in ankle kinematics affect knee joint loading, which can lead to individualized orthoses designs, adjustments during training or personalized injury recovery plans. More specifically, we want to find if certain ankle trajectory adaptations reduce the mechanical loading on the joint.

Methodology

Using the K3R CAMS dataset subject's morphology and kinematics as a reference [4], we generated several gait patterns, where we limited the ankle range of motion, using PredSim [5]. Next, we used the outputs from PredSim to build the specific files for the OpenSim analysis tools (Fig. 1). From these, we computed the static optimization values from which to obtain the force file, and finally completed the study on the knee joint load.

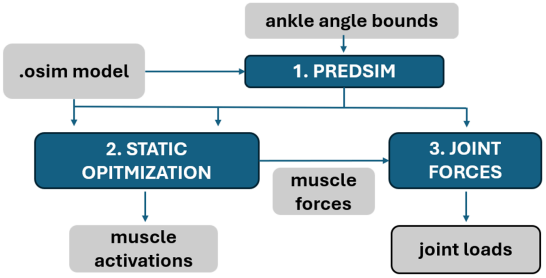


Figure 1: Outline on the procedure to obtain joint loading through OpenSim.

Results

Ankle trajectories with more extreme dorsiflexion-only patterns generated higher crouch behavior during gait (Fig. 2). While this pattern yielded lower peak ground reaction forces, net joint torques increased because the point of application of the force and the center of rotation moved apart in the transverse plane. These created bigger knee reaction forces, and therefore, higher cumulative knee joint loads, with higher ankle dorsiflexion (Fig. 3).

Overall, we found that dorsiflexion-only patterns could produce knee loads lower than in standard conditions (blue pattern in Fig. 3) when dorsiflexion was moderate, but produced increased knee loads when dorsiflexion was pronounced.

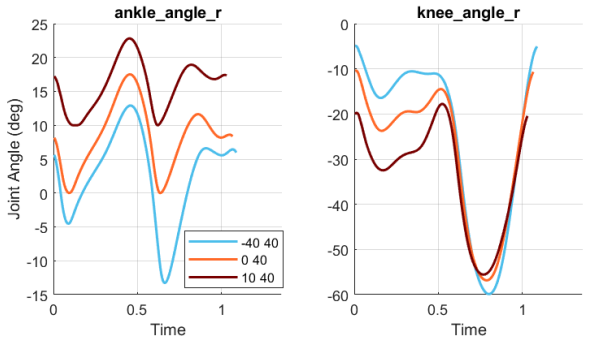


Figure 2: Right ankle and knee kinematics for the whole gait cycle for healthy gait (blue), neutral dorsiflexion (orange) and extreme dorsiflexion (brown) on lower bound of ankle kinematics.

Conclusions

We found a relationship between ankle kinematics and knee joint loading. To ensure the results generate an impact in patients, it should be confirmed experimentally and could be applied in recovery plans or orthoses designs when the ankle flexibility of the patient is very limited in order to force the joint to gradually adapt to a less straining movement.

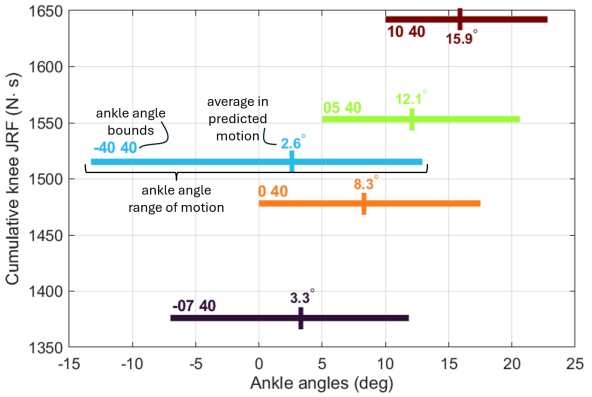


Figure 3: Overall cumulative knee joint load in a full gait cycle for each simulation. Every horizontal line represents the ankle angles ranges and their vertical placement the accumulation of the load that has been accumulated throughout the cycle.

References

[1] J. B. Morrison. Journal of Biomechanics, 1970
[2] Antoine Falisse. PLOS ONE, 2017
[3] Carmichael F. Ong. PLOS Computational Biology, 2019
[4] William R. Taylor. Journal of Biomechanics, 2017
[5] D'Hondt, L., et al. IEEE BioRob, 2024